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Research paper

Restoration of speech packet estimating residue of burg linear prediction

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ABSTRACT

Restoring speech packets is an inevitable phenomenon in Voice over Internet Protocol (VoIP) communications. This study proposes an accurate speech packet reconstruction for packet loss by estimating and incorporating the residual error of linear prediction (LP) into the actual linear prediction approach. Residual error refers to the difference between the actual speech signal and its predicted counterpart. The speech signal is non-stationary due to changing statistical properties such as frequency, amplitude, and spectral envelope over time, resulting in mismatches in frame-by-frame estimation if residual error is ignored. Ignoring residual error in conventional LP-based packet loss concealment (PLC) degrades reconstructed speech quality because it influences the system to generate a model-restricted and overprocessed speech signal. The result may seem numerically accurate, but it feels muffled, unnatural, and distorted in perception because the error component significantly affects realism. We employed the Burg LP method along with residual error estimates rather than traditional LP-based PLC methods. The conventional LP approaches minimize only the forward prediction error, whereas the Burg method minimizes the forward and backward prediction errors by applying the Levinson-Recursion (L-R) algorithm. We deployed the Burg linear prediction in the forward and backward directions to reconstruct the lost packet with higher accuracy. The Burg method effectively addresses numerical problems, creates a stable model, and delivers an accurate LP estimation even with short data lengths. However, in numerical analysis, the autocorrelation (ACR) method for LP encounters challenges in precise LP estimation. This study estimates the residual error in the packet loss case and integrates it into conventional Burg linear prediction estimates. Incorporating residual error estimation into Burg LP enhances reconstructed speech quality over previous methods because it influences statistically compatible, realistic, and perceptually natural speech signal restorations. The restoration method prevents overly confident estimations by accurately modeling the estimated residual error. Error integration to the base method minimizes spectral discontinuities. In addition, speech becomes less artificial and easier to understand when residual errors are distributed properly. The error-awareness method is more effective because it includes uncertainty. In this study, the simulation results from the proposed PLC method demonstrate significant performance improvements in both perceptual and objective evaluations compared to current LP-based PLC approaches.

1. Introduction

The massive use of internet telephony has expanded the opportunity for VoIP technology, which is used to operate internet phones [1]. In the case of Internet telephony, the cost of a call is comparatively lower than that of a conventional phone call [2]. The transmitted speech signal is inferior through a packet-switched network owing to issues such as additive noise, bit error, network congestion, and packet delay [1]. Thus, recipients receive degraded speech signals at the receiver's end. To upgrade the degraded quality of a speech in a speech communication system, the portion of the lost speech signal must be reconstructed. The process of recovering missing speech signal segments is called packet loss concealment (PLC). Waveform-based pitch waveform replication (PWR) and waveform replication (WR) approaches have been proposed for PLC techniques [2,3]. Moreover, LP-based recovery sub-codec (RSC) and modified covariance methods have been suggested for the PLC technique [4,5]. Among contemporary PLC techniques, deep neural networks (DNN) and hidden Markov models (HMM)

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are discussed as leading-edge techniques in modern PLC approaches [6–8]. However, an enormous speech data set is applied to train the DNN and HMM models, which causes a hands-on problem under several conditions. On the other hand, the PLC technique based on the linear prediction method leads to significant performance with reduced computational complexity. Only limited work has been discussed in the literature on the LP-based PLC method. Accordingly, LP-based PLC techniques have ample room to boost their performance. In conventional LP, a key issue is avoiding the residual error estimate and not integrating it into the actual LP estimate [4,5]. The speech signal is non-stationary due to changing statistical properties such as frequency, amplitude, and spectral envelope over time, resulting in mismatches in frame-by-frame estimation if residual error is ignored. Ignoring residual error in conventional LP-based packet loss concealment (PLC) degrades reconstructed speech quality because it influences the system to generate a model-restricted and overprocessed speech signal. The result may seem numerically accurate, but it feels muffled, unnatural, and distorted in perception because the error component significantly affects realism. The prevailing LP methods minimize only the forward prediction error by applying the L-R algorithm. In numerical analysis, the prevailing LP methods face a challenge in calculating a precise LP. To mitigate the above problems, we applied the Burg LP method along with residual error estimates instead of the prior LP-based PLC methods.

Aoki [3] suggested a PLC method using PWR on account of the pitch variation for the forward-backward frame of the lost packet. This 2-sided PWR improved speech quality compared to conventional methods. This PWR method requires alignment techniques because the length of the replaced pitch cycles does not align with the placement and frame loss lengths. Moreover, the pitch period is longer, and there is a high probability of signal discontinuity between the received and lost packets. According to [4], a recovery sub-codec (RSC) technique was exposed for PLC in which a low-delay code-excited LP (LD-CELP) was employed. The RSC method is a modified version of the autocorrelation method. In [5], the modified covariance (MC) method was executed from the least-squares system for the PLC technique in noise-robustness conditions, where bone-conducted (BC) speech was transmitted. This MC method for PLC investigated how the dispatched BC speech signal was restored with better precision by minimizing the dynamic range of the speech signal spectra. Qiang Ji et al. [9] recommended a PLC technique by combining two methods such as (i) the first one is built on DNN and waveform similarity overlap-add (WSOLA) and, (ii) the second one is founded on the DNN and Griffin–Lim algorithm (GLA). This combination of the PLC method resolved incorrect phase spectral estimation of the speech signal for the lost frame. In [10], a regularized modified covariance (RMC) based PLC technique was employed for BC speech. The RMC-based PLC method reconstructs speech signals with better precision by compressing the dynamic range of the speech signal spectra. In [11], a PLC approach was suggested to enhance the decoded speech grade of CELP-type speech coders according to burst packet loss circumstances for wireless sensor networks. In [12], a PCM-coded PLC method utilized a pitch period and error signal of the preceding packet to generate excitation for packet loss. Furthermore, a PCM-coded PLC technique is not capable of resolving speech signal degradation because backward packet prediction from the forthcoming frame is not considered at all.

In the literature, familiar PLC techniques are based on pitch waveform replication, linear prediction, HMM, and DNN models. According to the prevailing PLC techniques [3–12], none of the studies estimated and considered residual errors in the respective methods. WR and PWR are typically waveform-based PLC techniques in which waveform and pitch waveform are replicated. The waveform-based PLC method requires an alignment technique because the replaced pitch cycle length does not align with the location and packet loss length. On the other hand, autocorrelation, covariance, modified covariance, and Burg methods are commonly LP-based approaches. For short data lengths, the autocorrelation method is inappropriate [13]. The effectiveness of the ACR approach is influenced by the speech signal's spectral dynamic range. [14]. The application of the L-D algorithm with rectangular windowing beneath the ACR approach further exacerbates bias propagation [15,16]. Typically, a pre-emphasis technique is unused in the PLC process. The covariance method addresses residual errors solely in the forward direction, while the MC approach reduces them in both directions [17]. Thus, variance-based methods provide better results; however, they are unstable models of input speech data. Therefore, prevailing LP-based PLC methods are not always suitable for retrieving lost frames.

From the above viewpoints, we focus on a stable model, specifically the Burg method, where we estimate and integrate predictive residual errors. The Burg method always yields a stable model and high resolution for short data lengths [18]. In addition, for spectral estimation from short data records, the Burg method classically outperforms the Yule-Walker (Y-W) method [19]. This study emphasizes the estimation of the residual errors in the forward and backward frames. The estimated residual error is included in each packet loss estimation during lost frame reconstruction. Integrating residual error estimation into Burg LP improves the restored speech quality compared to prevailing methods because it specifies statistically compatible, realistic, and perceptually natural speech signal restorations. The restoration method prevents overconfident estimations by accurately modeling the estimated residual error. Error integration to the base method reduces spectral discontinuities. In addition, speech becomes less artificial and easier to understand when residual errors are distributed naturally. The error-awareness method is more effective because it includes uncertainty. Thus, the restored speech acquired better quality. AC speech is affected by the surrounding noise. In cases where noise robustness is crucial, LP analysis can be significantly impacted by ambient noise. This occurs because the LP technique generates noise-corrupted parameters in noise-robust cases. Consequently, these parameters deviate from those derived from clean speech, making it difficult to represent the original vocal tract information. Therefore, when a noisy AC speech signal is transmitted, the LP-based PLC methods experience degraded performance. Moreover, LP methods can lead to ill-conditioning in numerical analysis due to the large spectral dynamic range of the input speech signal. Thus, in this work, the Burg LP method is affected by noisy cases and broader spectral dynamic ranges. In the experiments, perceptual and objective evaluations were performed, and it was observed that the proposed PLC method was more suitable than conventional PLC techniques.

2. Proposed method

This study proposes a suitable speech restoration technique that emphasizes residual errors. Residual errors influence speech signal restoration, a significant phenomenon in enhancing prevailing LP methods.

2.1. Burg method

The autoregressive (AR) process is mathematically formulated as

$$z(k) = -\sum_{j=1}^{p} \boldsymbol{\alpha}(j).z(k-j) + e(k) \tag{1}$$

where $z(k)$ $(k = 0, 1, 2, \ldots\ldots)$ is a sampled signal; p and $\alpha(j)$ denote the model order and AR model coefficients, respectively; and $e(k)$ is the residual error. The Burg method estimates the reflection coefficients and then utilizes the Levinson-Recursion (L-R) algorithm to obtain the AR parameters. The reflection coefficients were acquired by recursively minimizing the prediction error powers [18,19]. Considering the reflection coefficients $(c_1, c_2, \ldots, c_p)$ estimates, the AR parameters are obtained as

$$r_{xx}(0) = \frac{1}{N}\sum_{k=0}^{N-1} |z(k)|^2 \tag{2}$$

$\alpha(1) = c_1$

$\rho_1 = (1 - |\alpha_1(1)|^2) r_{xx}(0)$

For $c = 2, 3, 4, \dots, p$,

$$\alpha_c(i) = \begin{cases} \alpha_{c-1}(i) + c_c \alpha_{c-1}(c-1) & for \quad i = 1, 2, \dots, c-1 \\ c_c & for \quad i = c \end{cases} \tag{3}$$

Therefore, $\rho_c = 1 - |\alpha_c(c)|^2 \rho_{c-1}$ (4)

$\alpha_p(1)$, $\alpha_p(2)$,..., $\alpha_p(p)$ and ρ_p correspond to AR parameter estimates and white noise variance. The AR parameters in the Burg method are estimated by reducing the forward and backward residual errors using the least-squares technique [20]. The forward and backward prediction errors are expressed as

$$e_p^f(k) = z(k) + \sum_{j=1}^{p} \alpha_p(j).z(k-j) \tag{5}$$

$$e_p^b(k) = z(k-p) + \sum_{j=1}^{p} \alpha_p(j).z(k-p+j) \tag{6}$$

where $e_p^f(k)$ and $e_p^b(k)$ are the forward and backward LP errors, respectively. However, the AR parameters are recurrently estimated using reflection coefficients. The reflection coefficients (c_p) are expressed as follows [20]:

$$c_p = \frac{-2 \sum_{k=p}^{N-1} e_{p-1}^f(k)\, e_{p-1}^b(k-1)}{\sum_{k=p}^{N-1} (\{e_{p-1}^b(k-1)\}^2 + \{e_{p-1}^f(k)\}^2)} \tag{7}$$

2.2. Residual error calculation

The prevailing LP methods have a common problem: a large segment loss results in a greater amplitude degradation of the recovered speech, which leads to a significant decline in speech quality. In this context, the proposed approach emphasizes the speech signal estimation model for lost frames and assumes a cause with attenuation. The prevailing speech estimation model for LP is represented as:

$$z(k) = -\sum_{j=1}^{p} \alpha(j)\, z(k-j) \tag{8}$$

The estimation model was expanded and the transition was accomplished. Therefore, (8) can be rewritten as (9) as follows:

$$z(k) = -\sum_{j=1}^{p} \alpha(j)\, z(k-j)$$

$$= -\alpha(1).z(k-1) - \alpha(2).z(k-2) - \cdots - \alpha(p).z(k-p)$$

$$z(k) + \alpha(1).z(k-1) + \cdots + \alpha(p).z(k-p) = 0 \tag{9}$$

(9) indicates a first-order linear combination of sample points in the linear prediction approach. By observing (1) and (9), the prevailing estimation model (9) considers the prediction error $e(k)$ to be zero (0). It is concluded that the amplitude of the acquired signal waveform is degraded owing to the absence of predictive residual errors in speech voice estimation. Therefore, the prediction error that occurs actually in the packet loss from adjacent packets must be estimated. Therefore, we incorporated prediction errors into the model of the speech signal waveform estimates to improve the execution of the prevailing LP method. The predictive residual error is obtained from the deviation between the actual and estimated speech waveforms. The linear prediction error was estimated in the frontward and behindhand directions for the adjoining packets. The predictive error $\hat{e}(k)$ in the missing segment is determined using the LP error $e(k)$ at the adjoining packets as follows:

$$\hat{e}(k) = z(k) - \hat{z}(k)\,; (k = 0, 1, \dots, L-1) \tag{10}$$

where L is the packet-loss length. The error estimation $\hat{e}(k)$ is beneficial because the original speech signal and LP error have the same periodicity. The LP error of the lost frame is estimated using the PWR technique because the signal periodicity is ascertained. Although the prevailing PWR approach processes speech signal waveforms, processing is accomplished for the LP errors in this case. Therefore, the input to the autocorrelation function changes to linear prediction errors as follows:

Algorithm 1. Residue calculation.

```
func error := residue (s, w, α, p)
    for j := p + 1 : w
        zEsti := −α^T × s(j − 1 to j − p)
    end for
    if size (zEsti, 1) ≠ size(s, 1)
        zEsti := zEsti^T
    end if
    error := s − zEsti
end func
```

$$R(m) = -\sum_{j=0}^{L-1} e(j)\, e(j-m) \tag{11}$$

Algorithm 1 represents the process of residue calculation. In Algorithm 1, w and s indicate the window length for pitch extraction and samples for error estimate, respectively, and α represents the AR coefficients of LP order p. T corresponds to the transpose of a matrix. Finally, the estimated linear prediction error was added to the prevailing estimation model to reconstruct the packet loss. As a method of incorporation, it takes the form of adding the linear prediction error as

$$\hat{z}(k) = -\sum_{j=1}^{p} \alpha(j)\, z(k-j) + \hat{e}(k) \tag{12}$$

where $z(k-j)$ and $\hat{z}(k)$ denote the preceding and predicted samples, respectively, and $\alpha(j)$ indicates the LP coefficients of LP order p.

3. Packet loss estimation

The generic LP equation with residual error incorporation (12) is expressed as

$$\hat{z}(k) = -\boldsymbol{\alpha}\mathbf{z}^T + \hat{e}(k) \tag{13}$$

where $\boldsymbol{\alpha}$ and $\mathbf{z}$ are defined as follows,

$$\boldsymbol{\alpha} = \{\alpha(1)\ \alpha(2)\ \dots\ \alpha(p-1)\ \alpha(p)\} \tag{14}$$

$$\mathbf{z} = \{z(k-1)\ z(k-2)\ \dots\ z(k-p)\} \tag{15}$$

$$\hat{e}(k) = e^f(k) + e^b(k) \tag{16}$$

where T indicates the transpose operation of a vector. $\boldsymbol{\alpha}$ and $\mathbf{z}$ indicate the LP parameter vector and the input data vector, respectively. The parameters in $\boldsymbol{\alpha}$ are estimated from past and forthcoming frames to packet loss using Burg linear prediction. The elements in (14) and (15) are expressed more clearly and specifically as follows:

$$\boldsymbol{\alpha}(1:p) = \boldsymbol{\alpha} \tag{17}$$

$$\mathbf{z}(k-1:k-p) = \mathbf{z} \tag{18}$$

Using (17) and (18), (13) is represented as

$$\hat{z}(k) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k-1:k-p) + \hat{e}(k) \tag{19}$$

In the forward and backward predictions, (19) is performed L times to determine data samples in lost packets when packet loss holds L missing data samples. Burg linear prediction is deployed in forward-backward prediction to minimize the predicted speech signal attenuation. Forward and backward estimations are accomplished from the prior and forthcoming frames to packet loss. In the forward estimates, the missing samples were obtained as follows,

Algorithm 2. Addition of estimated packets.

```
func z := add (FEP, BEP)
    L := length(FEP)
    z := zeros([L 1])
    for j := 1 to L
        Ω := (j − 1)/(L − 1)
        z(n + j) := FEP(j) × (1 − Ω) + BEP(j) × Ω
    end for
end func
```

$$z^f(1) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k:k-p+1) + e(k)$$
$$z^f(2) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k+1:k+1-p+1) + e(k+1)$$
$$\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$
$$z^f(L) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k+L-1:k+L-1-p+1) + e(k+L-1)$$
$$z^f(L) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k+L-1:k+L-p) + e(k+L-1) \tag{20}$$

where L indicates a lost packet's length. Note that k in (20) denotes the number of the last received samples in the preceding packet to the lost packet. Furthermore, in the case of backward estimates, the n is revised as $k = k + L + 1$. Thus, k indicates the first number of samples of the succeeding frame of the lost packet. In backward prediction, missing samples are determined as follows:

$$z^b(1) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k:k+p-1) + e(k)$$
$$z^b(2) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k-1:k-1+p-1) + e(k-1)$$
$$\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$
$$z^b(L) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k-L+1:k-L+1+p-1) + e(k-L+1)$$
$$z^b(L) = -\boldsymbol{\alpha}(1:p)\ \mathbf{z}^T(k-L+1:k-L+p) + e(k-L+1) \tag{21}$$

3.1. Estimated frames addition

The addition operation is performed between forward and backward predicted frames after being stored in the buffer. The packet loss rate is reduced because the buffering delay increases and vice versa [2]. For this reason, the voice-over-Internet protocol is deployed by setting a jitter buffer to eschew jitter delay [21]. Data samples obtained from forward and backward estimates were multiplied by weighted values. The forward and backward predicted frames received from the jitter buffer are summed to compensate for missing packets. Algorithm 2 represents the addition of the preceding and subsequent frames where FEP and BEP correspond to the forward packet estimate and backward packet estimate, respectively. Also, L represents the complete sample used for measuring packet loss. We can define linear weighting Ω as $0 \le \Omega \le 1$. We applied linear gain to avoid weakening the speech signal. Fig. 1 denotes an illustrative layout of a 2-sided Burg linear prediction, where the predicted residue is included in forward and backward estimations. The forward prediction is performed from the previous frame of the lost packet, and the behind-hand estimate is from the forthcoming frame shown in Fig. 1 (c) (Algorithm 1 is deployed). Thereafter, the speech is restored by summing forward and backward predicted frames as shown in Fig. 1 (d) (Algorithm 2 is deployed). The proposed method takes the benefits of linear prediction through the forward and backward adjoining packets to the packet loss so that the estimated speech cannot be distorted. The proposed method overcomes the drawback of the estimated speech's attenuated amplitude.

4. Experiments

In this study, ITU-T G.711 μ-law quantization and 8 kHz sampling frequency were used, according to the recommendation by G.711 [22]. We used air-conducted (AC) speeches obtained through an AC microphone and digital recorder. The input speeches were approximately 4 seconds in length. Input speeches were from different sentences pronounced by 20 male and 20 female speakers. The assessment outcome of each experiment was 100 epochs and averaged because the evaluation results can vary between speakers, even when the packet loss rate remains constant. As an objective evaluation, we considered the perceptual evaluation of speech quality (PESQ) to estimate the fineness of the reconstructed speech. PESQ measures speech quality in a range from -0.5 to 4.5. A large scale signifies better speech quality [23]. In this paper, we considered the transmitted speech as the reference signal and the restored speech to be evaluated. Furthermore, log area ratio (LAR), log-spectrum distortion (LSD), and spectral bias (SB) were employed as objective evaluations. The simulation conditions are shown in Table 1.

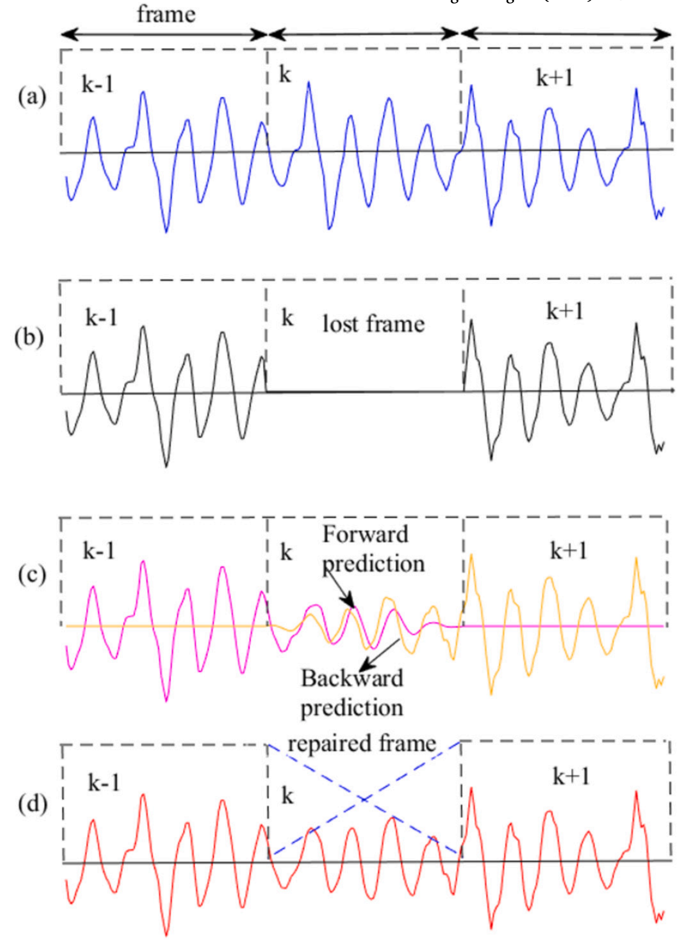

Fig. 1. Illustrative layout of 2-sided Burg linear prediction: (a) True speech signal, (b) Distorted speech signal, (c) Forward-backward estimates, (d) Repaired speech signal.

Table 1
Experimental conditions [2,23].

Consideration	Characteristics
Speech	Nihongo
Speaker	Male and female
Single speech signal length	4 s
Sampling frequency	8 kHz
Quantization	ITU-T G.711 μ-law
Windowing	Rectangular
Packet length	10 ms
Packet loss rate	10% & 30%
Number of trials	100 times
LP analyzing packet length	20 ms
LP order	12
Evaluation through	PESQ, LSD, LAR, SB

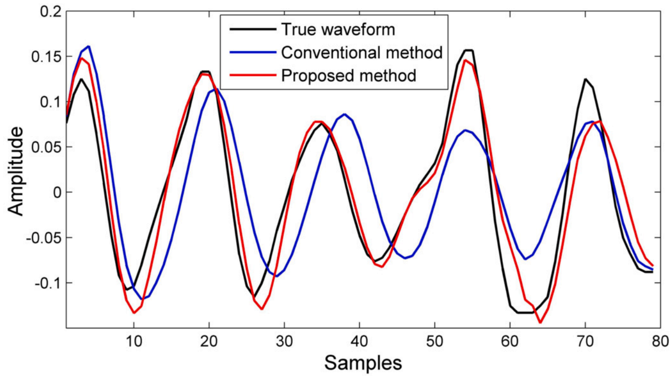

Fig. 2. Estimated waveforms.

Table 2
PESQ scores (average) of some PLC methods.

Methods	10% loss rate		30% loss rate	
	male	female	male	female
ACR	3.66	3.70	3.21	3.28
RSC	3.75	3.78	3.32	3.36
MC	3.74	3.79	3.36	3.39
RMC	3.84	3.87	3.43	3.47
Burg	3.87	3.90	3.48	3.50
Proposed	**4.12**	**4.16**	**3.72**	**3.75**

4.1. Speech dataset

This study implements air-conducted (AC) speech recorded from the concept of the NRIPS database [24]. We measured the AC speech using a SONY MODEL ECM-16 AC microphone and recorded it using a ROLAND R-44 digital recorder. In the AC speech instance, we used 10 diverse sentences pronounced by 20 male and 20 female speakers; therefore, we deployed 200 male and 200 female pronunciations of the AC speech for the simulation.

5. Results

The results were inspected through perceptual and objective evaluation of AC speech. In the Result section, we compare the prevailing and proposed PLC techniques.

5.1. Perceptual evaluation

The visual representation of the reconstructed speech signal was considered a perceptual assessment. The estimated waveform of the restored speech signal was assumed to be a visual representation. Fig. 2 shows the waveform estimated using the conventional and proposed methods. The prevailing PLC technique applies a linear gain [4,5,25,26], while the proposed method reconstructs the speech signal without applying a linear gain. Fig. 2 shows the restored speech waveforms obtained from the prevailing method (avoiding error estimation) and the proposed method (with error estimation). The proposed PLC technique reconstructs the speech waveform that is very similar to the actual dispatched speech signal. In this context, the inclusion of estimated error signals upgrades the restored speech signals' condition.

5.2. PESQ scores

The proposed LP-based PLC technique abstains from the usage of linear gain in reconstructed speech signals. On the other hand, prevailing LP-based PLC techniques deploy a linear gain in the reconstructed speech signal [2,5,26]. Table 2 lists the PESQ scores of the different PLC techniques. In Table 2, the Burg linear prediction obtained acceptable PESQ scores for the AC speech. However, the proposed method (Burg method with error estimates) acquires the best PESQ scores in contrast to the prevailing PLC techniques. The prevailing Burg linear prediction obtains 3.87 and 3.90 PESQ scores in 10% packet loss for males and females, respectively. In contrast, the proposed Burg linear prediction acquires 4.12 and 4.16 PESQ scores in 10% packet loss for male and female speakers, respectively. Likewise, the proposed PLC technique attains acceptable PESQ scores with 30% packet loss for male and female speakers. We conclude that the proposed PLC technique obtains an acceptable PESQ score because the residual error is included in the prevailing Burg linear prediction method. For a fair comparison, we used a common input dataset for all the employed PLC techniques.

5.3. Statistical analysis

5.3.1. Confidence intervals

Confidence intervals (CIs) measure the statistical reliability of PESQ scores. CI reflects the uncertainty of the mean score of PESQ and confirms that significant improvements in PLC are not due to random variation in the test samples. From Table 2, in 10% packet loss for the male speaker, the average PESQ of the baseline method (Burg), $M_1 = 3.87$, and the average PESQ of the proposed method ($Burg + Integrated\_Error$), $M_2 = 4.12$. We estimated the standard deviation of the baseline method and the proposed method, corresponding to $SD_1 = 0.13$ and $SD_2 = 0.16$, respectively, as shown in Table 3. We counted the total of male speech $n = 200$ utterances for both PLC methods. CI is estimated as follows:

$$CI = \Delta M \pm t_{\alpha,df} \times SE \tag{22}$$

where $\Delta M(M_2 - M_1 = 0.25)$ is the mean difference of PESQ between baseline PLC and proposed PLC. SE denotes the standard error. For a 95% confidence interval (CI), we know the notation t_α, df, where α is a significance level and df (n-1 = 199) is the degrees of freedom. In a two-sided 95% CI, the middle 95% of the distribution is kept, leaving 5% split into two tails (2.5% in each). The significance level α is estimated as follows:

$$\alpha = 1 - \frac{0.05}{2} = 0.975 \tag{23}$$

Therefore, the notation is written as $t_{0.975,199}$ (t-value), and the value of this notation is calculated using the following code segment:

```
from scipy.stats import t
t.ppf(0.975, 199)
```

From the above code segment, we obtain the value of $t_{0.975,199} = 1.972$. However, the standard error (SE) is estimated as,

$$SE = \sqrt{\frac{SD_1^2 + SD_2^2 - 2r * SD_1 * SD_2}{n}} \tag{24}$$

Therefore,

$$SE = 0.01458;\ [plausible\ correlations\ r = 0,\ without\ pair\ information]$$

From (22), we compute CI as follows:

$$CI = 0.25 \pm t_{0.975,199} \times SE$$
$$= 0.25 \pm 1.972 \times 0.01458$$
$$CI = 0.25 \pm 0.02875$$

Therefore, confidence interval ≈ [0.221, 0.279]. CI does not include 0, hence the proposed PLC's mean PESQ (4.12) is significantly higher than the baseline mean PESQ (3.87). Table 3 corresponds to the standard deviation and mean difference (ΔM) of PESQ for Baseline PLC and Proposed PLC. Table 4 shows the CI of 10% and 30% packet loss for male and female speakers. In Table 4, all intervals exclude 0; therefore, the increases ($\Delta M = 0.25, 0.26, 0.24, 0.25$) are statistically significant.

Table 3
Standard deviation and mean difference of PESQ for Baseline PLC and Proposed PLC.

Methods	10% loss		30% loss	
	male	female	male	female
Baseline PLC (Burg)	0.13	0.13	0.14	0.12
Proposed PLC (Burg + error)	0.16	0.14	0.17	0.15
Mean difference of the PESQ (Proposed PLC-Baseline PLC)	0.25	0.26	0.24	0.25

Table 4
CI of male and female speech in different loss rates with r = 0 and 0.5.

Statistical analysis	10% loss		30% loss	
	male	female	male	female
CI (no pair information, r = 0)	[0.221, 0.289]	[0.233, 0.287]	[0.209, 0.271]	[0.223, 0.277]
CI (pair information, r = 0.5)	[0.229, 0.271]	[0.241, 0.279]	[0.218, 0.262]	[0.231, 0.269]

Table 5
Pooled standard deviation (PSD) and Effect Size of male and female speech in different loss rates.

Statistical analysis	10% loss		30% loss	
	male	female	male	female
PSD	0.146	0.135	0.156	0.136
Effect Size	1.71	1.93	1.54	1.84

Table 6
Degrees of freedom, t-statistic, t-value, and p-value of male and female speakers.

Significant testing	10% loss		30% loss	
	male	female	male	female
Degrees of freedom (df)	398	396	384	379
t- statistic	17.12	19.26	15.42	18.41
t-value	1.96594	1.96597	1.96616	1.96624
p-value	2.9×10^{-65}	1.45×10^{-82}	5.2×10^{-54}	1.0×10^{-75}

5.3.2. Effect size

In the PLC study, the 'effect size' measures the level of enhancement or degradation that a PLC technique provides compared to a baseline approach. The effect size estimates the degree of effect for the practical impact on speech intelligibility and quality. The 'effect size' depends on both the mean difference and the standard deviation (SD). Cohen's d score measures the effect size. We determined Cohen's d by calculating the mean difference of the PESQ score between the baseline and proposed PLC methods, then dividing the output by the pooled standard deviation (PSD). The 'effect size' is estimated by Cohen's d formula as follows:

$$Cohen's\ d = \frac{Average\ PESQ\ of\ proposed\ PLC - Average\ PESQ\ of\ baseline\ PLC}{Pooled\ SD} \tag{25}$$

In addition, the pooled standard deviation is estimated as follows:

$$Pooled\ SD = \frac{SD\ of\ PESQ\ for\ proposed\ PLC + SD\ of\ PESQ\ for\ baseline\ PLC}{2} \tag{26}$$

Table 5 corresponds to the 'effect size'. In this section, we consider the Burg-based PLC as the baseline method and the error-integrated Burg method as a proposed PLC technique. In Table 5, we observe that the effect sizes of 10% packet loss are 1.71 and 1.93 for male and female speakers, respectively. For packet losses of 30%, the effect sizes 1.54 and 1.84 correspond to male and female speakers, respectively. Cohen's d provides benchmarks for effect sizes of 0.2, 0.5, and 0.8 as small, medium, and large, respectively. The large 'effect size' indicates an improvement of the proposed PLC method in practice as shown in Table 5.

5.3.3. Significant testing

In this section, we consider the p-value and t-statistic for significance testing. The p-value is less than alpha (often $\alpha = 0.05$), the result is statistically significant, and the null hypothesis is rejected. The hypothesis of no correlation or effect refers to the null hypothesis. The p-value is the valid evidence of whether the null hypothesis is rejected or not. The p-value is calculated by the complementary error function (erfc) as follows:

$$p = erfc\left(\frac{|t|}{\sqrt{2}}\right) \tag{27}$$

where t denotes the $t - statistic$ value. The t-statistic is estimated as

$$t = \frac{\Delta M}{SE} \tag{28}$$

where ΔM and SE denote mean difference and standard error, respectively. SE is estimated using (24). We estimated the degrees of freedom using the Welch–Satterthwaite formula as

$$df \approx \frac{\left(\frac{SD_1^2}{n_1} + \frac{SD_2^2}{n_2}\right)^2}{\frac{\left(\frac{SD_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{SD_2^2}{n_2}\right)^2}{n_2 - 1}} \tag{29}$$

Table 6 shows degrees of freedom, t-statistic, t-value, and p-value of male and female speakers. The effect on the p-value is negligible because df is large. df is important for small samples, resulting in a higher p-value. In Table 6, $|t - statistic| > t - value$, the null hypothesis is rejected. The t-value is calculated from the code segment presented in Section 5.3.1. In Table 6, we noted that the p-value is extremely small for each observation. Therefore, the observed PESQ improvements (0.25, 0.26, 0.24, and 0.25) are highly statistically significant.

5.4. Log spectrum distortion

The LSD is the distance between the actual and estimated spectra of a speech signal. The average LSD scores for the reconstructed speech obtained from some PLC techniques are listed in Table 7. We consider a frame length of 20 ms with a 10 ms frameshift for the LSD calculation. For a single speech signal, the LSD of 20 frames is averaged by applying 10 ms frame overlapping. Log spectral distortion is defined as follows:

$$LSD = \sqrt{\frac{1}{B}\sum_{b=1}^{B} |20\ \log_{10}(\frac{P(\omega)}{\hat{P}(\omega)})|^2} \tag{30}$$

Table 7
LSD scores (average) of some PLC methods.

Methods	Speakers	
	Male	Female
ACR	12.01	11.98
RSC	11.79	11.53
MC	11.27	10.83
RMC	10.67	10.17
Burg	10.13	9.86
Proposed	**8.17**	**8.04**

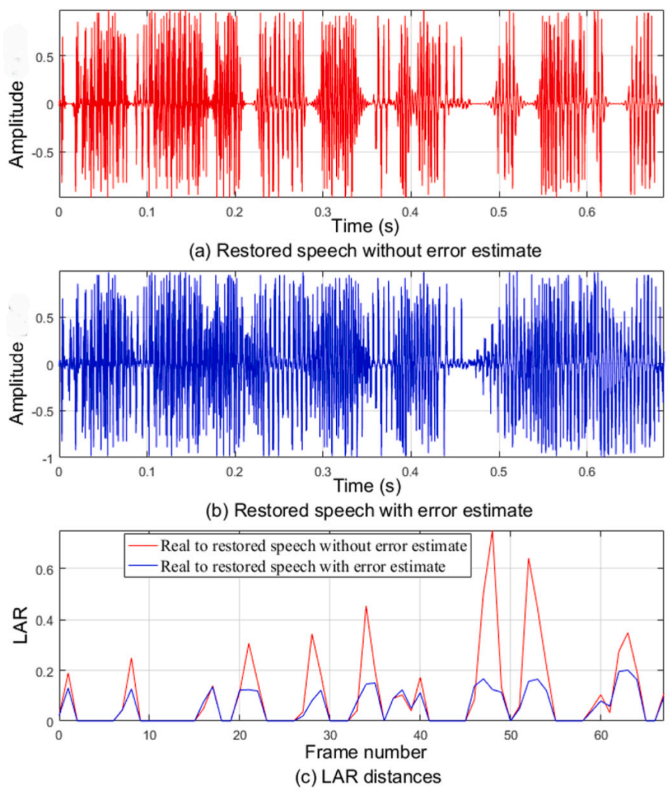

Fig. 3. Reconstructed speech waveforms and LAR distances: (a) Restoration speech without error estimate, (b) Restoration speech with error estimate, (c) Log area ratio distances.

where $P(\omega)$ and $\hat{P}(\omega)$ are the spectra of the transmitted and restored speech signals, respectively. The number of upper-frequency bins is represented by B. As shown in Table 7, the proposed Burg linear prediction PLC method provided a lower LSD score than the prevailing PLC techniques. Restored speech is very close to transmitted speech because its LSD score is small [27]. Therefore, the proposed PLC technique compensates for packet loss with a higher accuracy than the prevailing PLC techniques.

5.5. Log area ratio

The log area ratio refers to the parameter employed to present the LP coefficient that is transmitted over a channel [28]. The log area ratio distances are shown in Fig. 3, where the restored speech signal with a residual error estimate yields smaller values than the conventional Burg method of LP. A small LAR score denotes a well-reconstructed speech signal that approximates the true speech signal [29]. In Fig. 3, the LAR distance line of the proposed method is near the baseline, in contrast to that of the conventional method; therefore, the transmitted speech is precisely reconstructed in the packet loss through the proposed Burg linear prediction PLC method. The proposed PLC technique results in an LAR distance that remains close to the baseline, specifying better speech reconstruction.

Table 8
Average spectral biases of restored speech.

Restored Speech	Methods					
	ACR	RSC	MC	RMC	Burg	Proposed
M_S1	0.571	0.349	0.309	0.290	0.267	0.172
M_S2	0.468	0.312	0.239	0.220	0.197	0.147
M_S3	0.502	0.417	0.341	0.318	0.293	0.196
M_S4	0.476	0.374	0.328	0.305	0.278	0.187
M_S5	0.623	0.423	0.390	0.366	0.337	0.227
Average	**0.528**	**0.375**	**0.321**	**0.300**	**0.274**	**0.186**
F_S1	0.604	0.382	0.353	0.328	0.300	0.205
F_S2	0.501	0.345	0.275	0.253	0.230	0.180
F_S3	0.535	0.450	0.380	0.357	0.326	0.229
F_S4	0.509	0.407	0.353	0.334	0.311	0.220
F_S5	0.656	0.456	0.426	0.399	0.370	0.260
Average	**0.561**	**0.408**	**0.356**	**0.334**	**0.307**	**0.219**

5.6. Spectral bias

The spectral bias of the restored speech signal obtained from different methods was investigated using conventional and proposed methods. The spectral bias was calculated as [30,31]

$$SB = \frac{2}{f_s} \int_0^{f_s/2} \frac{|\hat{P}(\omega) - P(\omega)|}{P(\omega)} d\omega \tag{31}$$

where $P(\omega)$ and $\hat{P}(\omega)$ are the spectra of the transmitted and restored speech signals, respectively. Table 8 lists the spectral biases of different methods. In Table 8, the proposed method leads to the least spectral bias compared to conventional methods; thus, the proposed method reconstructs lost packets more precisely than existing PLC methods.

6. Conclusion

We recommend an LP-based speech reconstruction technique in which the predictive residual error is estimated and included in the restored speech signal. In this study, Burg's linear prediction was considered as a linear prediction approach. The calculated predictive error is included in the forward and backward predictions to improve the attenuation of a speech signal. Burg linear prediction is deployed to ensure a stable model for reliable output estimation because the speech signal is non-stationary. The proposed Burg linear prediction did not utilize a linear gain at all; commonly, prior PLC techniques used a linear gain to avoid signal attenuation. Experiments in perceptual and objective evaluations achieved results using the proposed PLC technique, which is superior to the prevailing PLC techniques. Thus, the proposed Burg linear prediction method improves the prevailing LP-based PLC techniques. The key findings and their importance are demonstrated as follows:

- Key finding: Modeling residual error improves the naturalness of restored speech
- Importance: This error modeling reduces small artifacts in the prevailing PLC, resulting in more natural speech
- Key finding: Error modeling PLC improves objective quality (PESQ, LSD, LAR, SB) and perceptual realism
- Importance: The objective and perceptual evaluations confirm that the restored speech is more relevant to the actual one
- Key finding: By integrating the remaining error, better adjustment to network jitter and burst loss is achieved
- Importance: Real-time robustness communication under poor networks
- Key finding: Smooth transition when a packet restarts due to residual error estimates

- Importance: Mitigates irritating “clicks” when a true packet arrives after a loss

In this study, the integration of an estimated surplus error with the obtained output signal indicates the inclusion of the input signal. Future research will integrate linear and polynomial predictions to accelerate the PLC technique, rather than relying on the estimation and addition of error signals. Thus, we can reduce residual error by generating a new PLC technique rather than directly mitigating the error signal. For multimodal VoIP communication, we will consider both audio and video data for further study in PLC techniques.

CRediT authorship contribution statement

Ohidujjaman: Writing – review & editing, Writing – original draft, Methodology, Conceptualization, Formal analysis. **Mahmudul Hasan:** Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Suman Ahmmed:** Writing – review & editing, Validation, Formal analysis, Data curation. **Mohammad Nurul Huda:** Visualization, Supervision, Investigation. **Mohammad Shorif Uddin:** Visualization, Supervision, Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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